Foundations of Data Science for Biologists

Pandas

BIO 724D

2026-MAR-31

Instructors: Greg Wray and Kayla Wilhoit

Learning objectives for today

Become familiar with panda's **Series** and **DataFrame** data structures

Learn how to **create**, **modify**, **index**, **query**, and **sort** ndarrays

Apply these skills to manipulating and analyzing images

Feedback on final project proposals

A top-level summary can be very helpful (flowchart or similar)

Think about where you will be running analyses (laptop, cluster)

Identify any steps that require special hardware (GPUs, large RAM, etc.)

Identify the software you will need for each step (analysis, formatting, transfer, etc.)

Make note of file formats: source and transformed data, intermediate, archival, etc.

Think about how much data you will be analyzing, including intermediate files

Think about where the data will reside and whether it will need to move

Describe the foundational outputs: tables, figures, images, etc.

Include a plan for archiving data and code: repositories, any discipline-specific features

Introduction to pandas

pandas

pandas is the most widely used dataframe library for Python

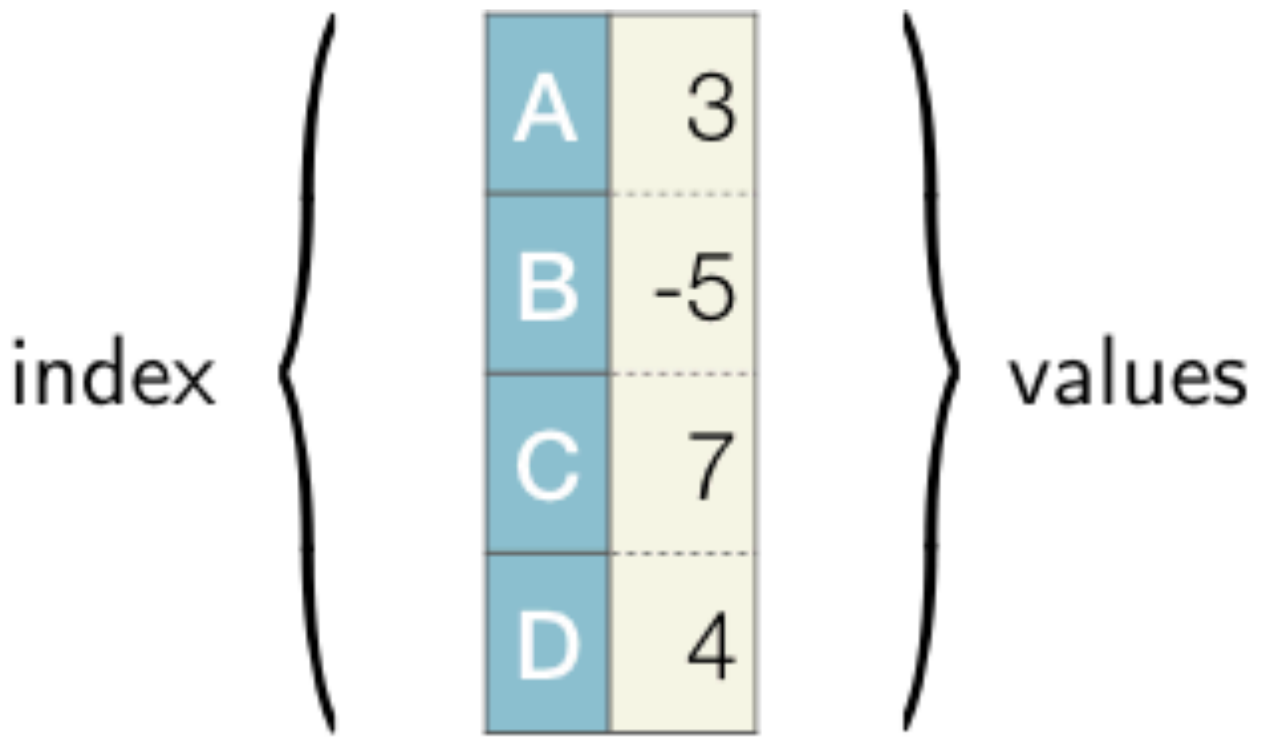
Provides two fundamental data structures

- **DataFrame**: analogous to data frames and Tibbles in R
- **Series**: effectively a single column of a DataFrame
- Both have a **row index**: integer by default but can also be strings

Provides many methods and attributes for working with tabular data

- Analogs of the functions provided by the Tidyverse set of libraries in R

pandas data structures

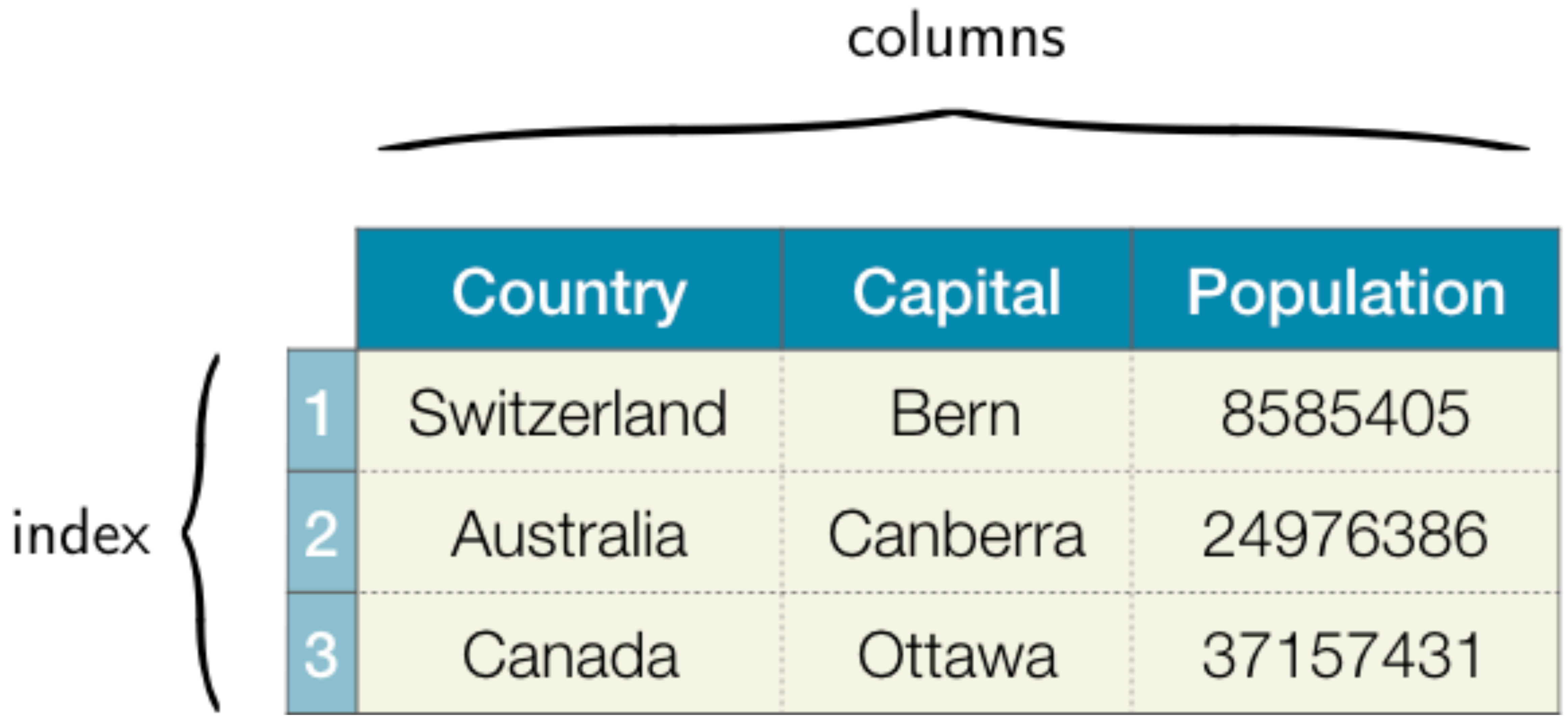


Series

1D labelled array

Values can be any data type

Values must all be the same data type



Series

2D labelled array

Columns can be any data type

Columns can differ in data type

Values in a column must be same type